Functional summary

Honeybee head: proteins and RNA

MOFA | 20 matched samples | Proteomics + RNA | Workspace 53df3b0f

Prepared from the completed browser run. 4 components; top 20 features per sign and view.

Feature alignment: matched_genes. Feature scaling: pareto. Center: on. Whole-view scaling: on. Maximum missingness: 30%. Seed: 557.

Component overview

Factor 1 | Metabolism and biosynthesis

Proteomics: 33.7% reconstruction | RNA: 1.8% reconstruction

0 term tests pass report-wide BH q < 0.05. Candidate.

Factor 2 | Gene-expression and biosynthetic regulation

Proteomics: 5.3% reconstruction | RNA: 28.0% reconstruction

30 term tests pass report-wide BH q < 0.05. BH-supported.

Factor 3 | Metabolism and biosynthesis

Proteomics: 1.0% reconstruction | RNA: 30.5% reconstruction

1 term tests pass report-wide BH q < 0.05. BH-supported.

Factor 4 | Transport and cell organization

Proteomics: 18.1% reconstruction | RNA: 1.0% reconstruction

0 term tests pass report-wide BH q < 0.05. Candidate.

Reconstruction basis: centered variation

Reading the report

BH-supported themes contain a term with report-wide BH q < 0.05. Candidate themes are ranked functional leads. Each factor has at most one interpretation supported by enrichment and reviewed primary studies. Positive and negative weights describe opposite ends of a component; the sign can flip between fitted models.

Factor atlas

MOFA | 20 matched samples | 4 components

Bars show reconstructed centered variation within each molecular view.
BH: Benjamini-Hochberg procedure. q: smallest report-wide adjusted p value in the leading functional theme. BH-supported: q < 0.05; candidate: q >= 0.05.
Source: current completed analysis; top 20 features per sign and view.

COMPONENT PROFILE

Factor 1: Metabolism and biosynthesis

Functional themes

Metabolism and biosynthesis | Candidate

cellular catabolic process. Score 1.62; 3 related terms; lowest report BH q 0.32.

cellular catabolic process [GO:0044248]: Proteomics, negative; hits 3/8; report q=0.32.

catabolic process [GO:0009056]: Proteomics, negative; hits 3/8; report q=0.54.

IDs: LOC408358, LOC411557, LOC412843

Stress and immune response | Candidate

response to stress. Score 1.63; 3 related terms; lowest report BH q 0.54.

response to stress [GO:0006950]: Proteomics, negative; hits 3/8; report q=0.54.

response to external stimulus [GO:0009605]: RNA, negative; hits 3/9; report q=0.64.

IDs: GeneID_412484, GeneID_727161, LOC409979, LOC412843, LOC552285

Functional annotation profile | Candidate

regulation of biological process. Score 1.30; 3 related terms; lowest report BH q 0.87.

regulation of biological process [GO:0050789]: Proteomics, positive; hits 6/10; report q=0.87.

regulation of cellular process [GO:0050794]: Proteomics, positive; hits 6/10; report q=0.87.

IDs: LOC100576993, LOC100577697, LOC550655, LOC551123, LOC552844

Sample associations

group: Cranberry = 1: rho=-0.68, n=20; rna_library_size: rho=-0.18, n=20; province: Quebec = 1: rho=0.03, n=20

Supporting features

Leading measured features

Proteomics: LOC100576993 (1.287), LOC726721 (1.084), LOC410824 (0.897), LOC410058 (-0.891)

RNA: Dscam (0.260), Obp4 (0.227), LOC408358 (0.194), LOC552285 (-0.190)

COMPONENT PROFILE

Factor 2: Gene-expression and biosynthetic regulation

Functional themes

Gene-expression and biosynthetic regulation | BH-supported

regulation of biosynthetic process. Score 4.85; 18 related terms; lowest report BH q 0.000064.

regulation of biosynthetic process [GO:0009889]: RNA, positive; hits 10/16; report q=0.000064.

regulation of cellular biosynthetic process [GO:0031326]: RNA, positive; hits 10/16; report q=0.000064.

IDs: LOC100576251, LOC100576993, LOC408877, LOC409456, LOC409928

RNA processing and gene regulation | BH-supported

RNA metabolic process. Score 3.27; 18 related terms; lowest report BH q 0.0066.

RNA metabolic process [GO:0016070]: RNA, positive; hits 7/16; report q=0.0066.

regulation of RNA metabolic process [GO:0051252]: RNA, positive; hits 7/16; report q=0.0066.

IDs: LOC100576993, LOC408877, LOC409456, LOC409928, LOC411557

Functional annotation profile | BH-supported

regulation of biological process. Score 2.63; 6 related terms; lowest report BH q 0.016.

regulation of biological process [GO:0050789]: RNA, positive; hits 11/16; report q=0.016.

regulation of cellular process [GO:0050794]: RNA, positive; hits 11/16; report q=0.016.

IDs: LOC100576251, LOC100576993, LOC408576, LOC408877, LOC409456

Sample associations

year: rho=0.82, n=20; province: Quebec = 1: rho=-0.40, n=20; rna_TMM_factor: rho=0.35, n=20

Main interpretation and evidence

Gene-expression and biosynthetic regulation

Factor 2 captures an RNA expression pattern enriched for regulation of biosynthetic process. Its scores are associated with collection year (Spearman rho 0.82).

Data: regulation of biosynthetic process; RNA contribution 28.0%; 10 selected genes; report-wide BH q 6.38e-05. Genes: LOC100576251, LOC100576993, LOC408877, LOC409456, LOC409928, LOC411557, LOC412161, LOC551123.

[1] The source study identifies biosynthesis and protein-synthesis modules in matched honeybee RNA and protein measurements.

[8] Independent honeybee brain data identify coordinated gene-regulatory programs across cell types and behavioral maturation.

[10] Single-cell brain profiles resolve cell-type and caste-associated RNA expression, providing context for cellular composition in bulk head samples.

Validation: Check whether the same regulatory-gene pattern recurs in independent head pools, with crop and collection year balanced.

Leading measured features

Proteomics: LOC113218789 (-0.313), LOC725432 (-0.297), Obp18 (-0.264), LOC413117 (0.257)

RNA: LOC411557 (1.057), LOC408740 (0.992), LOC551123 (0.982), LOC412161 (0.968)

COMPONENT PROFILE

Factor 3: Metabolism and biosynthesis

Functional themes

Metabolism and biosynthesis | BH-supported

small molecule metabolic process. Score 2.38; 6 related terms; lowest report BH q 0.016.

small molecule metabolic process [GO:0044281]: RNA, negative; hits 6/10; report q=0.016.

carboxylic acid metabolic process [GO:0019752]: RNA, negative; hits 4/10; report q=0.12.

IDs: LOC100576895, LOC411188, LOC412166, LOC412843, LOC552249

Nutrient and small-molecule metabolism | Candidate

carbohydrate metabolic process. Score 2.55; 1 related terms; lowest report BH q 0.12.

carbohydrate metabolic process [GO:0005975]: RNA, negative; hits 4/10; report q=0.12.

IDs: GeneID_406131, LOC406114, LOC412843, LOC552249

Cell communication and adhesion | Candidate

cell adhesion. Score 2.10; 1 related terms; lowest report BH q 0.32.

cell adhesion [GO:0007155]: RNA, positive; hits 3/8; report q=0.32.

IDs: LOC409722, LOC410824, LOC412968

Sample associations

rna_TMM_factor: rho=0.45, n=20; year: rho=-0.29, n=20; province: Quebec = 1: rho=0.29, n=20

Supporting features

Leading measured features

Proteomics: LOC551123 (0.134), LOC552401 (-0.100), LOC100577376 (0.064), LOC408807 (-0.062)

RNA: CPF2 (1.762), LOC107965367 (1.690), LOC410038 (1.438), LOC551323 (1.218)

COMPONENT PROFILE

Factor 4: Transport and cell organization

Functional themes

Transport and cell organization | Candidate

transport. Score 1.13; 2 related terms; lowest report BH q 1.0.

transport [GO:0006810]: Proteomics, negative; hits 3/10; report q=1.0.

establishment of localization in cell [GO:0051649]: Proteomics, negative; hits 2/10; report q=1.0.

IDs: LOC409722, LOC724386, LOC726721

Metabolism and biosynthesis | Candidate

cellular biosynthetic process. Score 1.01; 1 related terms; lowest report BH q 1.0.

cellular biosynthetic process [GO:0044249]: Proteomics, positive; hits 5/11; report q=1.0.

IDs: LOC100576251, LOC409456, LOC552106, LOC724704, LOC724951

Sample associations

province: Quebec = 1: rho=0.52, n=20; group: Cranberry = 1: rho=0.50, n=20; rna_TMM_factor: rho=0.45, n=20

Supporting features

Leading measured features

Proteomics: LOC552401 (0.972), CSP3 (0.967), Def1 (0.874), LOC725380 (0.855)

RNA: Dscam (-0.169), LOC724607 (0.092), LOC724704 (0.088), LOC100576270 (0.078)

Reading by factor

Primary research is curated by study, tissue, functional theme and measured features. Interpretation support is labelled separately from functional context.

Factor 1 | Metabolism and biosynthesis

[15] FEMS Microbes, 2026 | Functional context

Tests propolis exposure and redox-gene expression in fat body and head glands, alongside gut microbiota.

[9] Journal of Advanced Research, 2026 | Functional context

Combines brain and gut proteomics with metabolomics across worker ages, including amino-acid and energy pathways.

[14] Insects, 2026 | Functional context

Measures stage-specific brain proteome responses to carbon dioxide, including clearance, antioxidant and amino-acid pathways.

[11] Journal of Experimental Biology, 2024 | Functional context

Links stress-induced glycine N-methyltransferase transcription with midgut sarcosine measurements and cellular methylation metabolism.

[16] Insects, 2026 | Functional context

Tests dietary effects on metabolic and immune-gene expression before and after acute heat exposure in adult workers.

[12] Scientific Reports, 2021 | Functional context

Tests small heat-shock-protein responses to several controlled disruptions of protein homeostasis.

[7] The FASEB Journal, 2023 | Functional context

Measures age-dependent flexibility of honeybee energetic metabolism; physiological context for metabolic annotations.

[5] Insects, 2024 | Functional context

Compares dietary protein sources using vitellogenin, abdominal lipid and hypopharyngeal-gland measurements.

Factor 2 | Gene-expression and biosynthetic regulation

[10] iScience, 2022 | Interpretation support

Single-cell brain profiles resolve cell-type and caste-associated RNA expression, providing context for cellular composition in bulk head samples.

[8] Nature Communications, 2025 | Interpretation support

Independent honeybee brain data identify coordinated gene-regulatory programs across cell types and behavioral maturation.

[1] Proteomics, 2026 (online 2025) | Interpretation support

The source study identifies biosynthesis and protein-synthesis modules in matched honeybee RNA and protein measurements.

[9] Journal of Advanced Research, 2026 | Functional context

Combines brain and gut proteomics with metabolomics across worker ages, including amino-acid and energy pathways.

[11] Journal of Experimental Biology, 2024 | Functional context

Links stress-induced glycine N-methyltransferase transcription with midgut sarcosine measurements and cellular methylation metabolism.

[16] Insects, 2026 | Functional context

Tests dietary effects on metabolic and immune-gene expression before and after acute heat exposure in adult workers.

Factor 3 | Metabolism and biosynthesis

[15] FEMS Microbes, 2026 | Functional context

Tests propolis exposure and redox-gene expression in fat body and head glands, alongside gut microbiota.

[9] Journal of Advanced Research, 2026 | Functional context

Combines brain and gut proteomics with metabolomics across worker ages, including amino-acid and energy pathways.

[10] iScience, 2022 | Functional context

Resolves brain cell types and caste-associated RNA expression; relevant to cellular composition in bulk head profiles.

[8] Nature Communications, 2025 | Functional context

Maps cell-type and spatial gene-regulatory programs in nurse and forager brains.

[14] Insects, 2026 | Functional context

Measures stage-specific brain proteome responses to carbon dioxide, including clearance, antioxidant and amino-acid pathways.

[11] Journal of Experimental Biology, 2024 | Functional context

Links stress-induced glycine N-methyltransferase transcription with midgut sarcosine measurements and cellular methylation metabolism.

[16] Insects, 2026 | Functional context

Tests dietary effects on metabolic and immune-gene expression before and after acute heat exposure in adult workers.

[7] The FASEB Journal, 2023 | Functional context

Measures age-dependent flexibility of honeybee energetic metabolism; physiological context for metabolic annotations.

Factor 4 | Transport and cell organization

[15] FEMS Microbes, 2026 | Functional context

Tests propolis exposure and redox-gene expression in fat body and head glands, alongside gut microbiota.

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Combines brain and gut proteomics with metabolomics across worker ages, including amino-acid and energy pathways.

[14] Insects, 2026 | Functional context

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[11] Journal of Experimental Biology, 2024 | Functional context

Links stress-induced glycine N-methyltransferase transcription with midgut sarcosine measurements and cellular methylation metabolism.

[16] Insects, 2026 | Functional context

Tests dietary effects on metabolic and immune-gene expression before and after acute heat exposure in adult workers.

[7] The FASEB Journal, 2023 | Functional context

Measures age-dependent flexibility of honeybee energetic metabolism; physiological context for metabolic annotations.

[5] Insects, 2024 | Functional context

Compares dietary protein sources using vitellogenin, abdominal lipid and hypopharyngeal-gland measurements.

Methods and reproducibility

Signed top-weight gene sets; one-sided hypergeometric over-representation. Within-query and report-wide Benjamini-Hochberg correction. Terms with at least two query hits are grouped using complete-linkage clustering at kappa >= 0.5, with at least three shared model genes. Cluster score is mean(-log10(raw term p)); it summarizes related terms and is not an additional p value. This is a DAVID-inspired local grouping, using ordinary hypergeometric p values rather than DAVID EASE scores.

Report-wide correction covers 1632 term tests across components, views, annotation catalogs and weight directions. Complete-linkage term clustering uses kappa >= 0.5, with at least three shared model genes. Positive/negative queries select up to 20 features each. Views contributing at least 1% reconstruction enter annotation queries.

Interpretation selection

At most one interpretation; report-wide BH q < 0.05, at least three selected supporting genes, at least 10% reconstruction in the supporting view, at least 16 samples, centered inputs and two explicitly reviewed relevant primary studies. The current reviewed interpretation covers specific RNA-regulatory annotations in honeybee head samples; other contexts show functional results. No single-gene or keyword-only citation fallback.

Settings

transform: auto

normalization: none

max_missing: 0.3

model_features: 100

standardize: true

feature_scaling: pareto

mofa_center: true

mofa_scale_views: true

factors: 4

iterations: 300

seed: 557

challenge: none

mask_fraction: 0.1

method: mofa

feature_alignment: matched_genes

References

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[4] The Gene Ontology knowledgebase in 2023. GENETICS, 2023.

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[6] Varroa destructor parasitism and Deformed wing virus infection in honey bees are linked to peroxisome-induced pathways. Proteomics, 2024.

https://doi.org/10.1002/pmic.202300312

[7] Age-related flexibility of energetic metabolism in the honey bee Apis mellifera. The FASEB Journal, 2023.

https://doi.org/10.1096/fj.202300654r

[8] Single-nucleus and spatial transcriptomics identify brain landscape of gene regulatory networks associated with behavioral maturation in honeybees. Nature Communications, 2025.

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[9] Comprehension of the age-dependent gut and brain interaction of honey bee workers by integration of multi omics approaches. Journal of Advanced Research, 2026.

https://doi.org/10.1016/j.jare.2025.07.045

[10] Single-cell transcriptomic analysis of honeybee brains identifies vitellogenin as caste differentiation-related factor. iScience, 2022.

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[11] Transcriptional control of a metabolic switch regulating cellular methylation reactions is part of a common response to stress in divergent bee species. Journal of Experimental Biology, 2024.

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[13] Comparative transcriptome analysis of adult worker bees under short-term heat stress. Frontiers in Ecology and Evolution, 2023.

https://doi.org/10.3389/fevo.2023.1099015

[14] Effects of Carbon Dioxide on Hemolymph and Brain Proteomes in Honey Bee Workers (Apis mellifera L.). Insects, 2026.

https://doi.org/10.3390/insects17060630

[15] A propolis-rich hive environment affects redox gene expression and gut microbiota at the individual and social level in honey bees. FEMS Microbes, 2026.

https://doi.org/10.1093/femsmc/xtag019

[16] Diet Quality Shapes Metabolic and Immune Transcriptional Responses of Honey Bees Under Acute Heat Stress. Insects, 2026.

https://doi.org/10.3390/insects17080780

Abbreviations and explanations

Pareto - Pareto scaling

Divides each feature by the square root of its standard deviation; centering is controlled separately.

rho - Spearman rank correlation coefficient

Describes how two quantities vary together by rank.

kappa - Cohen kappa similarity

Measures agreement between term-gene membership profiles beyond chance.

UniProt - Universal Protein Resource

A curated resource of protein sequences and functional annotations.

MOFA - Multi-Omics Factor Analysis

Finds latent factors that describe variation across several molecular layers.

MOFA2 - Multi-Omics Factor Analysis, version 2

The R software used to fit and inspect MOFA models.

PCA - Principal Component Analysis

Summarizes the main directions of variation in a matrix.

PC - Principal Component

One axis obtained from principal component analysis.

GO - Gene Ontology

A structured vocabulary describing gene functions and biological processes.

BP - Biological Process

The Gene Ontology branch describing biological activities.

GO_BP - Gene Ontology: Biological Process

The biological-process annotation catalog used here.

KEGG - Kyoto Encyclopedia of Genes and Genomes

A database linking genes and compounds to pathways.

ORA - Over-Representation Analysis

Tests whether an annotation occurs more often in a selected feature list than in its background.

DAVID - Database for Annotation, Visualization and Integrated Discovery

A resource for gene annotation, enrichment and grouping related functional terms.

EASE - Expression Analysis Systematic Explorer

The modified Fisher-test score used by DAVID.

BH - Benjamini-Hochberg procedure

Adjusts p values for multiple comparisons.

FDR - False Discovery Rate

The expected proportion of false discoveries among selected results under the testing procedure.

q - BH-adjusted p value in this lab

A p value adjusted with the Benjamini-Hochberg procedure.

p - p value

The probability of a result at least this extreme under the specified null model.

R2 - Coefficient of determination

Fraction of the model input reconstructed relative to a zero baseline; with centering on, it describes variation around the mean.

K - Number of latent factors

The chosen number of factors in the model.

SD - Standard Deviation

Measures the spread of a feature. Unit-variance scaling divides by SD; Pareto scaling divides by its square root.

QC - Quality Control

Checks measurement gaps, scales and sample patterns.

CPM - Counts Per Million

RNA counts divided by the effective library size, multiplied by one million.

TMM - Trimmed Mean of M-values

An RNA library-size normalization method that trims extreme log-ratios.

TPM - Transcripts Per Million

RNA abundance adjusted for transcript length and library composition.

TMT - Tandem Mass Tag

An isotope-labeling method for multiplexed protein quantification.

MS - Mass Spectrometry

Measures ions by mass-to-charge ratio.

LC - Liquid Chromatography

Separates compounds before measurement.

LC-MS - Liquid Chromatography-Mass Spectrometry

Combines compound separation with mass-spectrometric detection.

MS/MS - Tandem Mass Spectrometry

Fragments ions to help identify molecules.

m/z - Mass-to-charge ratio

The mass of an ion divided by its charge.

RT - Retention Time

The time at which a compound elutes from a chromatography column.

Abbreviations, continued

HBB - Highbush Blueberry

One crop environment in the honeybee study.

CRA - Cranberry

The other crop environment in the honeybee study.

CCLE - Cancer Cell Line Encyclopedia

A collection of molecular measurements from cancer cell-line models.

T2D - Type 2 Diabetes

The disease group in the human-islet dataset.

RRID - Research Resource Identifier

The source identifier used to match the islet donors.

LOC - Locus identifier prefix

An NCBI label for a gene locus; its number is an identifier rather than a functional name.

GeneID - NCBI Gene Identifier

A database identifier for a gene.

WGCNA - Weighted Gene Co-expression Network Analysis

Groups genes with correlated expression patterns.

HTSeq - High-Throughput Sequencing analysis framework

Software used to count RNA-sequencing reads.

RNA-seq - RNA Sequencing

Measures RNA by sequencing RNA-derived molecules.

ROS - Reactive Oxygen Species

Reactive oxygen-containing molecules involved in signaling and oxidative stress.

TCA - Tricarboxylic Acid cycle

A central metabolic pathway that oxidizes carbon fuels.

Dscam - Down syndrome cell adhesion molecule

A cell-adhesion gene with roles in neuronal wiring and extensive alternative splicing.